

Assignment Set 2 — Defense Sheet

Computational Finance (WI4151), TU Delft — Prof. Fang Fang

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Scope: pricing a European basket call $V(0) = e^{-rT} E^Q[\max(w_1 S_1(T) + w_2 S_2(T) - K, 0)]$ with $w_1 = w_2 = 0.5$, $T = 1$, $r = 0.02$, $K = 100$, by three methods: Q1 Monte Carlo, Q2 moment matching, Q3 the 2D COS method. Two market setups recur in every question: **(a)** both underlyings log-normal (GBM), $S_{10} = S_{20} = 100$, $\sigma_1 = 0.20$, $\sigma_2 = 0.30$, $\rho = 0.30$; **(b)** S_1 log-normal, S_2 a CIR / square-root process $dS_2 = rS_2dt + \sigma_2\sqrt{S_2}dW_2$, $\sigma_1 = 0.25$, $\sigma_2 = 1.5$, $\rho = 0.10$ in Q1 / $\rho = 0$ in Q2 and Q3.

NOTE: the assignment PDF could not be rendered in this environment (no *poppler*/*pypdf*), so the “task in my own words” below is reconstructed from the in-code comments and parameters. **TODO (Bruno):** open `Computational_Finance_Assignment_2 (2) (1).pdf` and confirm the exact wording of each sub-question (especially what convergence behaviour Q1 and Q3 ask you to demonstrate).

1 The task (in my own words)

Price a two-asset European **basket call** under the risk-neutral measure Q . The basket is $B_T = w_1 S_1(T) + w_2 S_2(T)$; the payoff is $\max(B_T - K, 0)$, discounted at e^{-rT} .

- **Q1 — Monte Carlo.** Simulate the basket and report the price with a standard error / 95% CI.
 - (a) both assets GBM, correlated, exact terminal simulation.
 - (b) S_1 GBM, S_2 a CIR square-root process under the Q -drift rS_2 ; demonstrate (i) the weak / time-discretization convergence of the Euler scheme and (ii) the statistical $O(1/\sqrt{M})$ convergence.

- **Q2 — Moment matching.** Compute the raw moments of B_T in closed form, then approximate the basket distribution by matching 2, 3, 4 moments (log-normal, shifted log-normal, Johnson SU) and price the call from each. Benchmark against a high-precision Monte Carlo price.
- **Q3 — 2D COS.** Price the basket with the two-dimensional COS method (Ruijter & Oosterlee 2012), expanding the joint density of the two state variables on $[a1, b1]x[a2, b2]$, exploiting $\rho = 0$ so the 2D density coefficients factorise into a product of 1D coefficients. Show convergence in N .

TODO (Bruno): confirm whether the PDF asks for these specific deliverables or also for, e.g., a comparison table across the three methods.

2 My approach and why

- **Q1:** exact GBM simulation at maturity where possible (no time grid needed); Euler–Maruyama with full truncation for the CIR leg because exact joint simulation is unavailable when $\rho \neq 0$.
- **Q2:** derive $E[B_T^k]$ analytically from the binomial expansion of $(w1S1 + w2S2)^k$ and the cross moments of correlated log-normals (and the non-central chi-square moments for the CIR leg), then fit successively richer distributions to more moments.
- **Q3:** 1D characteristic functions per marginal, cumulant-rule truncation range per marginal, payoff coefficients by 2D trapezoidal quadrature, assembled as $chi1^T V chi2$.

TODO (Bruno): for each method, state in one sentence *why* it is the right tool and what each buys you (MC = unbiased baseline, moment matching = cheap closed-form approximation, COS = spectral accuracy). Do not let the examiner hear a generic answer.

3 Key code sections (file + what it does + why I wrote it that way)

3.1 Q1 — basket Monte Carlo (Q1_basket_MC (1).py)

- **simulate_basket_lognormal(M)** — **lines 17–42.** Q1(a). Draws two standard normals ($Z1, Zc$, lines 27–28) and correlates them by Cholesky: $Z2 = \rho * Z1 + \sqrt{1 - \rho^2} * Zc$ (line 29). Simulates terminal prices with the **exact GBM solution** at T (lines 32–33), forms the basket and discounted payoff (lines 36–37), and returns the MC mean and standard error $\text{std}(\text{ddof}=1)/\sqrt{M}$ (lines 40–41).
- **Convergence loop** — **lines 48–63.** Prices over $M = 2000 \dots 1e6$ (line 48), then estimates the convergence rate as the OLS slope of $\log(SE)$ vs $\log(M)$ (line 62), expected -0.5 .
- **simulate_basket_cir(M, n_steps)** — **lines 93–125.** Q1(b). Time-steps with $h = T/n_steps$ (line 107). $S1$ exact GBM step (line 118); $S2$ **Euler–Maruyama** step with **full truncation** $S2p = \max(S2, 0)$ inside drift and diffusion (lines 120–121), so the square root is always defined.
- **Weak-error test** — **lines 134–185.** Reuses the **same fine Brownian increments** across all coarse grids (**basket_on_grid**, lines 144–159; increments summed into blocks at lines 151–152) to isolate the discretization bias from MC noise. Fits the weak order as the slope of $\log|bias|$ vs $\log h$ on the coarse grids (line 184), expected 1.0 .
- **Statistical-error test** — **lines 198–210.** Fixes the grid, varies M , fits the SE slope (line 209).

TODO (Bruno): justify (i) full truncation vs reflection / other CIR fixes; (ii) why you reuse the same increments across grids for the weak-error test; (iii) the Q-drift argument at lines 86–91 (physical CIR drift $\kappa p(\theta - S2)$ replaced by $rS2$ because the discounted price is a Q-martingale).

3.2 Q2 — moment matching (Q2_moment_matching (1).py)

- *lognormal_cross(j, m, ...)* — **lines 20–29**. Cross raw moment $E[S_1^j S_2^m]$ for two correlated log-normals, from $E[\exp(N)] = \exp(\text{mean} + 0.5\text{var})$ with the combined log-return mean/variance/cov at lines 26–29.
- *basket_moments_LN(...)* — **lines 31–37**. Raw basket moments $E[B_T^k]$ via the binomial expansion $\sum_j C(k, j) w_1^j w_2^{(k-j)} E[S_1^j S_2^{(k-j)}]$.
- *cir_S2_rawmoments(...)* — **lines 39–53**. Raw moments of $S_2(T) = c \cdot Z$, $Z \sim \chi_0^2(\text{lambda})$ (zero-dof non-central chi-square), using cumulants $\text{kappa}_n = 2^{(n-1)} n! \text{lambda}$ (line 49) and the raw-from-cumulant recursion (lines 50–52). c and lambda from the technical hint at lines 46–48.
- *basket_moments_CIR(...)* — **lines 55–64**. Combines the LN- S_1 and CIR- S_2 marginal moments under independence ($\rho = 0$), so joint moments factorise (line 63).
- *central_from_raw(mom)* — **lines 66–75**. Converts raw moments to mean, variance, skewness, kurtosis.
- *price2moments* — **lines 78–89**. Fits a log-normal matching m_1, m_2 ; Black's formula with $F = m_1$, total variance $s^2 = \log(m_2/m_1^2)$.
- *price3moments* — **lines 91–107**. Shifted log-normal $B = \text{gamma} + L$; inverts the log-normal skewness $(w + 2)\sqrt{w - 1} = g_1$ for w by *brentq* (line 99), recovers shift and shifted strike.
- *price4moments* — **lines 109–139**. Johnson SU; solves the 2×2 system for $(\text{delta}, \text{gamma})$ from skewness and kurtosis by *least_squares* with Gauss–Hermite moments of *sinh* (U_{moments} , lines 120–125; residual lines 127–132; solve line 134), then prices by Gauss–Hermite quadrature (line 139). GH nodes/weights set at lines 16–17.
- **MC benchmarks** — *mc_benchmark_LN* lines 144–159, *mc_benchmark_CIR* lines 161–180. Exact simulation with antithetic variates; the CIR benchmark samples the zero-dof non-central chi-square via a Poisson mixture of Gammas (lines 171–173).

TODO (Bruno): be ready to (i) derive the cross-moment formula at line 29 by hand; (ii) derive the non-central chi-square cumulants $\text{kappa}_n = 2^{(n-1)} n! \text{lambda}$; (iii) explain *why* more matched moments need not monotonically reduce the pricing error.

3.3 Q3 — basket COS (Q3_basket_COS (1).py)

- *chi_coeffs(phi, a, b, N)* — **lines 22–32**. 1D density cosine coefficients $\text{chi}_k = \text{Re}\{\phi(\omega_k) \exp(-i \omega_k a)\}$, $\omega_k = k \pi / (b-a)$ (lines 29–30), with the $k = 0$ term halved (line 31) to realise the primed summation.
- *payoff_coeffs_2d(...)* — **lines 34–54**. Payoff cosine coefficients V_{k_1, k_2} : builds the payoff matrix $P = \max(w_1 g_1(y_1) + w_2 g_2(y_2) - K, 0)$ on a dense grid (line 49) and contracts it with cosine/trapezoidal-weight matrices C_1, C_2 (lines 52–53) as $C_1^T P C_2$ (line 54). Trapezoidal weights at lines 45–46; normalisation $(2/(b_1-a_1))(2/(b_2-a_2))$ at line 54.
- *price2dCOS(...)* — **lines 56–63**. Assembles $e^{-rT} \text{chi}_1^T V \text{chi}_2$ (line 63).
- *cumulant_range(k1, k2, k4, L = 12)* — **lines 65–71**. **The truncation range.** Returns $[k_1 - \text{width}, k_1 + \text{width}]$ with $\text{width} = L * \sqrt{(|k_2| + \sqrt{|k_4|})}$ — **lines 70–71**. This is the COS-paper cumulant rule; $L = 12$ is the default.
- **Q3(a) characteristic functions** — **lines 79–80**. Normal log-price CFs $\phi(\text{om}) = \exp(i \text{om} \mu - 0.5 \sigma^2 T \text{om}^2)$ for each marginal; means at lines 76–77.
- **Q3(a) ranges** — **lines 82–83**. *cumulant_range*($\mu, \sigma^2 T, 0$) per marginal (a normal has $k_1 = \text{mean}$, $k_2 = \text{variance}$, $k_4 = 0$).
- **Q3(b) S_2 characteristic function** — *phi2b*, **lines 117–119**. CF of $S_2(T) = c \cdot Z$, $Z \sim \chi_0^2(\text{lambda})$: $\exp(\text{lambda} i z / (1 - 2i z))$ with $z = \text{om} \cdot c$ (lines 118–119); c_b, lam_b at

lines 114–116.

- **Q3(b) cumulants of S_2 — lines 122–124.** $k_1 = c \cdot \text{lambda}$, $k_2 = c^2 \cdot 4 \cdot \text{lambda}$, $k_4 = c^4 \cdot 192 \cdot \text{lambda}$ (from $\text{kappa}_n = 2^{(n-1)} n! \text{ lambda}$ scaled by c^n).
- **Q3(b) ranges — lines 125–127.** S_1 range `cumulant_range(mu1_b, sigma1^2 T, 0)` (line 125); S_2 range `cumulant_range(k1_S2, k2_S2, k4_S2, L=12)` (line 126), then **clamped to be non-negative** $a_{2,b} = \max(a_{2,b}, 0.0)$ (line 127) because $S_2 \geq 0$.
- **Payoff maps — lines 84–85, 128–129.** $g = \exp$ for log-price state variables; $g_{2,b} = \text{identity}$ in Q3(b) because the S_2 state variable is the *level*, not the log.
- **Convergence loops — lines 88–104 (a), 131–145 (b).** Compare COS prices over $N = 8 \dots 64$ against a converged $N = 256$ reference.

4 Design decisions I must justify

- **Cholesky correlation** ($Z_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_c$) — Q1 lines 29, 116; Q2 line 152. **TODO (Bruno):** why this is exactly the 2×2 Cholesky factor.
- **Full truncation** of the CIR scheme — Q1 lines 120–121. **TODO (Bruno):** why truncation (not reflection) and what bias it introduces.
- **Reusing fine Brownian increments across grids** for the weak-error test — Q1 lines 144–169. **TODO (Bruno):** why this cleanly isolates discretization bias from MC noise.
- **Antithetic variates** in the Q2 benchmarks — Q2 lines 154–158, 176–179. **TODO (Bruno):** why antithetics reduce variance here and when they fail.
- **Choice of approximating families** (LN / shifted-LN / Johnson SU) — Q2 lines 78–139. **TODO (Bruno):** why each family is the natural one for 2 / 3 / 4 matched moments.
- $\rho = 0$ **factorisation** of the 2D density coefficients — Q3 lines 13–20, 60–63. **TODO (Bruno):** show why independence makes $c_{\{k_1, k_2\}} = \chi_{\{k_1\}} \chi_{\{k_2\}}$ and what changes if $\rho \neq 0$.
- $L = 12$ **in the cumulant rule** — Q3 line 65, used at lines 82–83, 125–126. **TODO (Bruno):** justify $L = 12$ (the paper often uses 8–10); relate to the bug below.
- **Trapezoidal quadrature** for V_{k_1, k_2} instead of a closed form — Q3 lines 34–54. **TODO (Bruno):** why no analytic V_k here (basket payoff couples the two state variables), and how N_{grid} was chosen.

5 Results and what they mean

TODO (Bruno): run the three scripts and fill in the actual numbers: - Q1(a) price + 95% CI, and the fitted SE slope (should be ≈ -0.5 , line 63). - Q1(b) price + CI, the fitted weak order (≈ 1.0 , line 184), the fitted SE slope (line 209). - Q2(a) and Q2(b): basket mean/std/skew/kurtosis, MC benchmark, and the 2/3/4-moment prices with absolute errors (printed at lines 196–198, 214–216) — explain why error generally falls with more moments and any exception. - Q3(a)/(b): converged COS price and the error/ratio table (lines 98–104, 140–145) — explain whether convergence looks spectral (exponential) and how the ratios behave.

6 The hardest question Fang could ask here — and my answer

Most likely deep-dive (highest probability): the COS truncation range $[a, b]$. Fang co-invented COS; the cumulant truncation rule is her result, and this assignment had a real bug there. She will go straight to `Q3_basket_COS (1).py:65–71` (`cumulant_range`) and its call sites `:82–83 (Q3a)` and `:125–127 (Q3b)`.

What the code factually does: `cumulant_range` returns $[k_1 - L \cdot \sqrt{(|k_2| + \sqrt{(|k_4|)})}, k_1 + L \cdot \sqrt{(|k_2| + \sqrt{(|k_4|)})}]$.

$\sqrt{(|k4|)})$ (Q3_basket_COS (1).py:70-71), with $L = 12$. For a normal marginal $k4 = 0$, so the half-width is $L * \sqrt{\text{variance}}$ (:82-83). For the CIR leg, the half-width uses the non-central chi-square cumulants $k2 = 4 \ c^2 \ \text{lambda}$, $k4 = 192 \ c^4 \ \text{lambda}$ (:123-124), and the lower end is clamped to 0 (:127).

The bug (special focus): a too-small $c2$ made $[a, b]$ too narrow at short maturities. The half-width scales like $\sqrt{k2}$, and $k2$ is the variance term — for a normal marginal $k2 = \text{sigma}^2 T$ (:82-83, :125), which **shrinks with T** . At short maturity $c2$ is small, so $\text{width} = L * \sqrt{c2 + \sqrt{c4}}$ (Q3_basket_COS (1).py:70) collapses and $[a, b]$ becomes too narrow to contain the support of the density / payoff, truncating mass and corrupting the price. The fix lives at the range construction — the *cumulant_{range}* formula at :70 and the L value at :65 (and the non-negativity clamp at :127).

TODO (Bruno): the *reasoning* of the fix — fill this in yourself. Specifically: 1. What symptom did you see (wrong/unstable price, or error not converging) and at which T ? 2. What exactly did you change — raise L , add the $\sqrt{k4}$ term, widen the floor, or change how $c2$ is fed in for short T ? Point to the precise line you edited (:65 $L = 12$, :70 the width formula, or :127 the clamp). 3. Why $L * \sqrt{c2 + \sqrt{c4}}$ (and not just $L * \sqrt{c2}$) is the right safeguard when $c2$ is tiny but $c4 > 0$, and why a fixed wide interval is *not* a free lunch (more terms N needed for the same accuracy, since COS error depends on both truncation and series length).

This is the single question most worth over-preparing.

7 “Show me in your code where you do X” — anticipated spots

- “...where you compute the truncation range $[a, b]$.” Q3_basket_COS (1).py:65-71 (*cumulant_{range}*), called at :82-83 (Q3a) and :125-127 (Q3b, with the $\max(a2_b, 0)$ clamp at :127).
- “...where a small $c2$ makes $[a, b]$ too narrow, and how you fixed it.” Q3_basket_COS (1).py:70 ($\text{width} = L * \sqrt{\text{abs}(k2) + \sqrt{\text{abs}(k4)}}$) and :65 ($L = 12$). **TODO (Bruno):** name the line you edited.
- “...where you define the characteristic function.” Q3(a) :79-80; Q3(b) CIR leg ϕ_{2b} :117-119.
- “...where you build the cosine density coefficients χ_k .” χ_{coeffs} Q3_basket_COS (1).py:22-32 (note the $\chi[0] = 0.5$ primed-sum term at :31).
- “...where you build the payoff coefficients $V_{k1, k2}$.” payoff_coeffs_2d :34-54 (assembly :54).
- “...where the 2D price is assembled.” $\text{price}_{2d\text{COS}}$:56-63 ($\chi1 @ V @ \chi2$ at :63).
- “...where the cumulants of the CIR leg come from.” Q3_basket_COS (1).py:122-124; same moments in Q2_moment_matching (1).py:39-53.
- “...where you correlate the Brownian motions.” Q1_basket_MC (1).py:29 and :116.
- “...where you do the Euler step for $S2$.” Q1_basket_MC (1).py:120-121 (full truncation).
- “...where you isolate the discretization bias.” Q1_basket_MC (1).py:144-169.
- “...where you match 2 / 3 / 4 moments.” price_2moments :78-89, price_3moments :91-107, price_4moments :109-139 in Q2_moment_matching (1).py.
- “...where the MC benchmark uses antithetics.” Q2_moment_matching (1).py:154-158 and :176-179.